

Lab 06 - B: Tres en Raya

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```
type Tablero = [Char]
```

```
mostrar :: Tablero -> IO ()
```

```
mostrar t = putStr $ "\n "  
    ++ [t!!0] ++ " | " ++ [t!!1] ++ " | " ++ [t!!2] ++  
    "\n-----\n "  
    ++ [t!!3] ++ " | " ++ [t!!4] ++ " | " ++ [t!!5] ++  
    "\n-----\n "  
    ++ [t!!6] ++ " | " ++ [t!!7] ++ " | " ++ [t!!8] ++ "\n\n"
```

```
mover :: Tablero -> Int -> Char -> Tablero
```

```
mover (x:xs) 0 j = j : xs
```

```
mover (x:xs) n j = x : mover xs (n-1) j
```

```
ganador :: Tablero -> Char -> Bool
```

```
ganador t j =  
    let lineas = [[t!!0,t!!1,t!!2], [t!!3,t!!4,t!!5], [t!!6,t!!7,t!!8],  
        [t!!0,t!!3,t!!6], [t!!1,t!!4,t!!7], [t!!2,t!!5,t!!8],  
        [t!!0,t!!4,t!!8], [t!!2,t!!4,t!!6]]
```

```
    ganadores = map (\linea -> foldr (\v acc -> v == j && acc) True linea) lineas  
    in foldr (\x acc -> x || acc) False ganadores
```

```
empate :: Tablero -> Bool
```

```
empate t = null (filter (\c -> c /= 'X' && c /= 'O') t)
```

```
jugar :: Tablero -> Char -> IO ()
```

```
jugar tab jugador = do  
    putStr $ "Turno de " ++ [jugador] ++ ". Elige posicion (0-8): "  
    mostrar tab  
    entrada <- getLine
```

```
    let jugadaValida = filter (\c -> [c] == entrada && c /= 'X' && c /= 'O') tab
```

```
    if null jugadaValida
```

```
        then do  
            putStr "Movimiento invalido. Intenta de nuevo.\n"  
            jugar tab jugador
```

```
        else do
```

```
            let pos = foldr (\x acc -> foldr (\n _ -> n) 0 (filter (\n -> [tab !! n] == entrada) [0..8])) 0
```

```
            entrada
```

```
            let nTablero = mover tab pos jugador
```

```
            if ganador nTablero jugador
```

```

then mostrar nTablero >> putStr ("Gano " ++ [jugador] ++ "\n")
else if empate nTablero
    then mostrar nTablero >> putStr "Empate \n"
    else jugar nTablero (if jugador == 'X' then 'O' else 'X')

```

```
main :: IO ()
```

```
main = do
```

```
    putStr "TRES EN RAYA \n"
```

```
    jugar ['0','1','2','3','4','5','6','7','8'] 'X'
```

The screenshot shows the OnlineGDB web interface. On the left is a sidebar with navigation links: OnlineGDB, code. compile. run. debug. share., IDE, My Projects, Classroom (new), Learn Programming, Programming Questions, Sign Up, and Login. The main area displays the execution of a C++ program. The code is in a file named 'main.hs'. The output shows the game progress:

```

[2 of 2] Linking a.out
TRES EN RAYA
Turno de X. Elige posicion (0-8):
0 | 1 | 2
-----
3 | 4 | 5
-----
6 | 7 | 8
0
Turno de O. Elige posicion (0-8):
X | 1 | 2
-----
3 | 4 | 5
-----
6 | 7 | 8
2
Turno de X. Elige posicion (0-8):
X | 1 | 0
-----
3 | 4 | 5
-----
6 | 7 | 8
4
Turno de O. Elige posicion (0-8):
X | 1 | 0
-----
3 | X | 5
-----
6 | 7 | 8
5
Turno de X. Elige posicion (0-8):
X | 1 | 0
-----
3 | X | 0
-----
6 | 7 | 8
8
X | 1 | 0
-----
3 | X | 0
-----
6 | 7 | X
Gano X
...Program finished with exit code 0
Press ENTER to exit console.

```